

Technical Note

THAWING FROZEN TYZOR® TPT

DESCRIPTION

Dorf Ketal Tyzor® TPT freezes at 18 °C (64 °F), which is very close to room temperature, so it may become solid during shipment or storage. Because Tyzor® TPT does not expand while freezing, containers will not bulge or show signs that the product has frozen.

Tyzor® TPT is 100% tetraisopropyl titanate and mixing is not necessary to obtain a homogeneous product when thawing. If frozen, Tyzor® TPT can be melted or thawed by careful, controlled heating. Overheating beyond the recommended temperature or prolonged storage at the limit of the recommended thawing temperature can cause the product to yellow significantly.

Tyzor® TPT is adversely affected by atmospheric moisture. The product will react irreversibly with moisture to produce isopropyl alcohol and hydrous titanium dioxide.

CAUTION

The flash point of Tyzor® TPT can decrease to <-6.67 °C (20 °F) when exposed to moisture because isopropyl alcohol is liberated during the reaction of Tyzor® TPT with water.

Tyzor® TPT freezes at 18 °C (64 °F). Follow these guidelines when thawing:

- Keep drum bungs tightly closed. (The vapor pressure from Tyzor® TPT at 46 °C (120 °F) is 0.098mm Hg and the vapor pressure at 108 °C (226 °F) is 14mm Hg.)
- Limit the thawing temperature to 71 °C (160 °F).
- Minimize time near the higher limit.
- Roll the drum to speed melting and improve uniformity.
- Cool the container quickly and store at 20 °C - 38 °C (68 °F - 100.4 °F).
- Keep drum bungs tightly closed until the product is used.
- Store the container in a warm, 20 °C - 38 °C (68 °F - 100.4 °F), dry location to prevent freezing.



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